EMIX: A DATA AUGMENTATION METHOD FOR SPEECH EMOTION RECOGNITION

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ABSTRACT

In the last few years, many deep learning (DL) models have been developed to improve the accuracy of speech emotion recognition (SER). However, as SER datasets are generally small and insufficient due to their difficult and expensive collection, the DL models are prone to overfitting, so their performance is limited. In this paper, we introduce a novel data augmentation (DA) method for the SER problem, namely EMix, which is simple but effective. The method creates new data by mixing pairs of selective samples from the original data. The generated mixtures will be noisier or less ambiguous than their constructive ones. To verify the effectiveness of the proposed DA, we develop a transformer-based network for the SER task, and experiment with the two public datasets including IEMOCAP and Crema-D. The experimental results demonstrate the superiority of EMix over other DA methods. In comparison with state-of-the-art methods, our approach shows competitive performance.

Index Terms— Speech emotion recognition, data augmentation, EMix

1. INTRODUCTION

Emotions are inevitable in human communication, and they can be expressed in many forms including facial expressions, gestures, texts, emojis, and voices according to contact methods. Correctly understanding human emotions enables us to have appropriate responses. In spoken languages, a speech emotion recognition (SER) system has many applications in various areas such as mental healthcare, human-machine interaction. Emotional states in speech are often hard to correctly recognize as they can be affected by many factors such as various speaking styles, diverse races and ethnicities, and different ages and genders. Additionally, emotional perception is subjective - different people could feel differently about an utterance, so emotion labels in SER datasets are usually determined by major votes of several annotators [1], which makes the labels noisy and ambiguous. These reasons make SER to be a complex and fine-grained classification problem.

1.1. Related work

In recent years, with the dominance of deep learning (DL), most researchers have been focusing on developing deep neural network (DNN) models to improve the performance of SER. The networks have been getting more complex and better with different architectures such as CNN [2, 3], RNN [4], ensemble model [5], and transformer [6, 7]. In general, applying DL for the SER task has two main problems: (1) data scarcity and (2) label ambiguity.

In the data scarcity problem, SER datasets are often relatively small with respect to the number of speakers and data volume. DNN models can be easily overfitted on the training data, but have poor generalization on the test data. A direct approach to deal with this problem is data augmentation (DA) which enlarges the training data size by constructing new data samples. Some DA methods in SER includes speed perturbation [8], vocal tract length perturbation (VTLP) [4, 9], Copy-Paste [2], mixup and GAN [10]. Some works also introduce different approaches for the problem such as multi-task learning [6] which employs an auxiliary task to add more constraint on the network, and transfer learning [2] which leverages a network that was trained on a relevant task like speaker verification.

To deal with the label ambiguity problem, [12] proposes a meta learning method which is a dynamic label correction method. It learns a new corrected label distribution to fix inaccurate labels in the dataset, and learns to estimate the contribution of each sample to the training process. Similarly, progressive co-teaching [13] uses two networks to automatically identify the difficulty level of data and then learn features of data from simple to difficult.

1.2. Our contribution

In this paper, we present a simple but effective data augmentation method for the SER task, which we call EMix. The method creates new data samples by linear combinations of selective data pairs from the emotion dataset. It has two basic mixing schemes - mixing with *Neutral* samples, and mixing within the same emotion. The former works like the noise augmentation method where the *Neutral* samples are like background noise, while the latter arguably increases label certainty of its new generated samples. EMix is label-preserving, and has no hyperparameter that requires tuning, so it is easy to apply. To verify its effectiveness, we implement a transformer-based network for the SER task, and conduct experiments with two public datasets: IEMOCAP [1] and Crema-D [14]. Our experimental results show that with the same network and the same experimental settings, EMix outperforms all other augmentation methods. Additionally, our proposed approach shows its competitive performance compared to some existing state-of-the-art approaches.

2. DATA AUGMENTATION

Data augmentation (DA) creates new data samples by applying constrained transformations on the original data. It diversifies the training data, and helps to reduce generalization gap between the training and test data. In this section, before introducing our proposed augmentation method for the SER task - EMix, we present Mixup [11] - a DA method which has demonstrated its effectiveness in many tasks, and it is the source of inspiration for our method.

2.1. Mixup

Mixup is a simple and data-agnostic augmentation method which constructs new virtual examples by linearly interpolating pairs of original examples. Given a pair of data samples (x_i, y_i) and (x_j, y_j) - where x_i, x_j and y_i, y_j are input vectors and labels, respectively, a new sample is derived by

$$\begin{aligned} \tilde{x} &= \lambda x_i + (1-\lambda)x_j \\ \tilde{y} &= \lambda y_i + (1-\lambda)y_j \end{aligned} \tag{1}$$

where $\lambda \sim Beta(\alpha, \alpha)$, for $\alpha \in (0, \infty)$.
As in the Eq.(1), mixup is not label-preserving. For implementation, the loss function when training a neural network with mixup is computed as

$$\tilde{L} = \lambda L(h(\tilde{x}), y_i) + (1-\lambda)L(h(\tilde{x}), y_j) \tag{2}$$

where L is a loss function that is applied on the original data, $h(x)$ is output of the neural network with input x.
Despite its broad applications, mixup has two drawbacks: (1) it requires more training time to achieve a good performance and (2) tuning the hyperparameter α to gain appropriate capacity is difficult and time-consuming [15]. In SER, the emotional state of an utterance does not necessarily spread all over the speech, it can sparsely concentrate on some segments. When mixing two utterances with different expressions, the emotional segments of them can be non-overlapped, so they are mostly intact in the augmented mixture. Thus, the mixture can clearly deliver two different emotions. These cases could harden the learning process of a network with the use of mixup for the SER task.

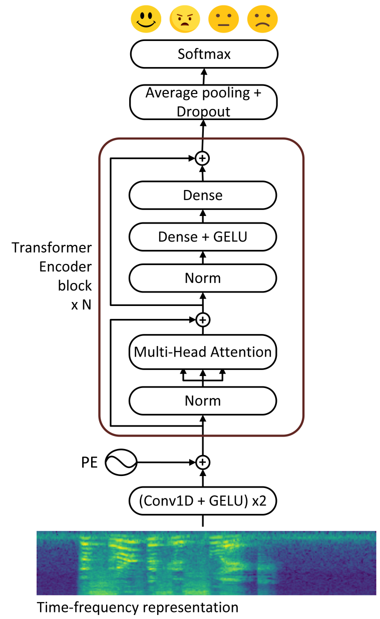

Fig. 1: The network architecture for speech emotion recognition.

2.2. EMix

Like mixup, our augmentation method for speech emotion recognition, which is EMix, also relies on convex combinations of data pairs to generate new samples, but different from mixup, the pairs are selectively chosen. According to its data selection, EMix has three mixing schemes: (i) mixing with *Neutral* emotion (EMix-N), (ii) mixing within the same emotion (EMix-S), and (iii) combination of the two formers (EMix-NS).

In EMix-N, utterances with any expression will keep their original emotional states when combining with *Neutral* utterances. Given a sample (x^e, y^e) with emotion e and a *Neutral* sample (x^n, y^n), a new sample is given by

$$\begin{aligned} \tilde{x} &= \lambda x^e + (1-\lambda)x^n \\ \tilde{y} &= y^e \end{aligned} \tag{3}$$

where $\lambda \sim U(0.5, 1)$ to preserve more information from the emotional sample. EMix-N can be seen as a case of noise augmentation where *Neutral* samples work like background noise.

In EMix-S, we only fuse speech recordings that have the same emotional states; the mixtures would have the same labels as that of their constructive recordings. As the presence of segments with the same emotion label increases in the mixture samples, the new data samples are considered to be more certain than that of the original constructive data. Thus, Emix-S is expected to work well with SER data that is noisy and ambiguous. It can be expressed as

$$\begin{aligned} \tilde{x} &= \lambda x_i^e + (1-\lambda)x_j^e \\ \tilde{y} &= y^e \end{aligned} \tag{4}$$

where $(\tilde{x}, \tilde{y})$ is the new sample, $(x^e_{i,j}, y^e_{i,j})$ are two data samples of emotion e, and $\lambda \sim U(0, 1)$.

Generally, EMix is label-preserving, so we can apply the same loss function to train on new generated samples. Additionally, it does not have any hyperparamter for tuning, which makes it easy to use, and saves training time.

3. NETWORK ARCHITECTURE

In order to verify the competence of the proposed DA method, we empirically develop a transformer-based network for the task of SER as the transformer [16] has shown compelling results on many sequential modeling problems. The network architecture is illustrated in Fig.1. It has input of a time-frequency representation which is fed to a stem block of two convolution layers followed by the GELU activation function [17] before mixing with positional embeddings and being processed by a stack of transformer encoder blocks. Output from the stack will be averagely aggregated, and randomly dropped. The softmax layer outputs probabilities of emotions.

The transformer encoder block is the main component of the network. Different from [16], we utilize pre-norm residual connection for its construction, which is known to be more efficient for back-propagation [18]. The block has a core module- multi-head attention (MHA), which allows for attending to different positions of the sequential input. MHA consists of h independent attention modules. Each module i processes the input in parallel by linearly transforming the input into query Q_i, key K_i, and value V_i before applying scaled dot-product attention to get attention head H_i. The h output attention heads are then concatenated and linearly transformed to produce the attentive output O. The processing steps are given as

$$
\begin{aligned}
&Q_i, K_i, V_i = XW_i^{Q,K,V}, i = \{1, ..., h\} \\
&H_i = softmax(\frac{Q_i K_i^T}{\sqrt{d/h}})V_i \\
&O = Concat(H_1, ..., H_h)W^O
\end{aligned} \tag{5}
$$

where $X \in \mathbb{R}^{l \times d}$ is the sequential input with length l and dimension d; the learnable weight matrices $W_i^{Q,K,V} \in \mathbb{R}^{d \times \frac{d}{h}}$, $W^O \in \mathbb{R}^{d \times d}$; and O is the sequential output.

4. EXPERIMENTS

4.1. Datasets

To validate the proposed method on the speech emotion recognition task, we conduct experiments with the two following datasets.

IEMOCAP [1]. The dataset includes improvised and scripted multimodal dyadic conversations between speakers (5 males and 5 females). It has five sessions with non-overlapping speakers, and consists of approximately 12 hours of speech. All recordings are judged by three annotators to determine their emotion labels. The labels are highly noisy and ambiguous because the dataset has 81.88% of recordings with more than one annotated expressions. In our experiments, we only use audio tracks of the improvised set, and classify four categorical emotions including *Angry*, *Happy*, *Neutral*, and *Sad*.

Table 1: Performance of our model with different data augmentation methods on IEMOCAP and Crema-D.

Augmentation method	**IEMOCAP**		**Crema-D**	
	WA	**UA**	**WA**	**UA**
No augmentation	72.59	65.31	74.28	74.08
Speed perturbation	74.39	67.60	74.98	74.72
VTLP	73.55	66.72	74.63	74.53
CopyPaste (1.5-seconds)	74.17	68.64	72.32	71.94
CopyPaste (3-seconds)	71.36	65.19	70.27	69.80
Mixup ($\alpha = 1.0$)	75.22	67.91	73.96	73.81
EMix-N	76.93	70.29	76.58	76.53
EMix-S	**77.63**	**71.85**	75.96	75.85
EMix-NS	77.32	71.03	**77.25**	**77.26**

Crema-D [14]. This is a multimodal dataset that has 7442 original clips from 91 actors (48 males and 43 females) with diversity in ethnicities and ages. The actors speak from a predefined set of 12 sentences in six different expressions (*Anger*, *Disgust*, *Fear*, *Happy*, *Neutral* and *Sad*) and four emotional levels (*Low*, *Medium*, *High* and *Unspecified*). Because of the professional actors, we can suppose utterances of Crema-D to be cleaner and less ambiguous than that of IEMOCAP. We choose the same set of four basic emotions like in IEMOCAP to experiment with this dataset.

4.2. Experimental Setup

4.2.1. Data preparation

We resample all speech recordings to 16 KHz, and split them into 3-seconds segments, then we extract 80-dimensional log-Mel magnitude spectrogram with a frame length of 25 ms and 10 ms step, which results in input features with the size of (301×80). All features are normalized to zero mean and unit variance through the time dimension with global mean and standard deviation being computed over all input features in the training set.

We randomly split each dataset into three sets of (train, validation, and test) with the ratio of 60/20/20. Following the common experimental settings of previous works, data splits in IEMOCAP are speaker-dependent while those in Crema-D are speaker-independent. We perform five-fold cross-validation in our experiments.

4.2.2. Network and training details

For fair comparison, we use the same network configuration for all experiments with the two databases. The network has input size of (301×80); the two convolution layers have 128 channels, kernel size of 3, and stride 1. Sinusoidal function is used for positional embeddings, and LayerNorm is deployed for all normalization layers. There are 6 transformer encoder blocks; they have an identical structure with input dimension of 128, 8 attention heads in MHA, and 512 and 128 units in the two dense layers, respectively. The dropout rate is fixed to 0.3.

To train the network, we utilize the cross-entropy objective function. For optimization, Adam optimizer with a starting learning rate of 0.0005 is our choice, and early stopping is also adopted in training processes. The optimal set of parameters is chosen from the best results on the validation set.

4.2.3. *Baselines*

To assess the efficacy of the proposed augmentation method, we experiment with different DA methods that are commonly applied for SER as our baselines. The methods include speed perturbation [8], vocal tract length perturbation (VTLP) [4, 9], mixup [11], and CopyPaste [2]. The two former methods transform an utterance to create a new one, while the two latter ones combine pairs of data samples. For speed perturbation, we resample audio signals to change their speeds in range of $[0.8, 1.2]$. VTLP bases on a speaker normalization technique to reduce interspeaker variability; its augmented value is in range of $(0.9, 1.1)$. CopyPaste concatenates speech segments within the same emotion or with *Neutral* segments, which leads to new longer samples. We try CopyPaste with segments of different lengths (1.5-seconds and 3-seconds), which corresponds to input lengths of 3 seconds and 6 seconds respectively. For mixup, a grid search is applied to find the best value of its hyperparameter α. Overall, the network architecture and experimental settings are the same for all DA methods.

4.3. Experimental results

We use weighted accuracy (WA), unweighted accuracy (UA) to report experimental results. WA is the overall classification accuracy of the entire test utterances, and UA is the average recall over the emotion categories.

The results obtained by the proposed transformer-based network with different DA methods on the two experimental datasets are shown in Table 1. EMix with its different schemes constantly outperforms all other augmentation methods in the two databases in both terms of WA and UA by large margins. Specifically, EMix-S achieves the best results on IEMOCAP with $WA = 77.63$ and $UA = 71.85$, which illustrates its effectiveness to work on noisy and ambiguous data like IEMOCAP. Additionally, combining EMix-S with EMix-N in EMix-NS somehow degrades the performance in this dataset. In contrast, EMix-NS with WA of 77.25 and UA of 77.26 has higher accuracies than EMix-N and Emix-S in Crema-D which is a cleaner and less ambiguous dataset. Moreover, while DA methods, that solely apply self-transformation on utterances like speed perturbation and VTLP, continually improve performance in comparison with no augmentation; combination-based DA methods like CopyPaste and mixup have unstable results over the two datasets. Mixup may need more time for training as we utilize early stopping. For CopyPaste, on one hand it makes inputs longer (6-seconds length) in case of 3-seconds constructive segments, so the network can be overfitted quickly with these long inputs; but on the other hand 1.5-seconds segments may not always contain emotional information of their original recordings.

Furthermore, we compare performance of our SER system with state-of-the-art methods. The comparison results can be seen in Table 2 and Table 3. Our model achieves the best WA value of 77.63, and ranks second in term of UA on IEMOCAP, while it outperforms others on Crema-D in terms of WA, UA and F-score.

Table 2: Performance comparison with state-of-the-art methods on the IEMOCAP dataset

Method	WA	UA
DDAE + Transfer learning [19]	70.07	70.67
TFCNN + DenseCap + ELM [3]	70.34	70.78
TRILL (10-folds) [20]	-	71.56
Transformer + Multi-task learning [6]	76.4	70.1
Head fusion + Noise augmentation[7]	76.18	**76.36**
Ours (*EMix-S*)	**77.63**	71.85

Table 3: Performance comparison with state-of-the-art methods on the Crema-D dataset

Method	WA	UA	F-score
ResNet-18 + SPEL [5]	-	68.12	-
SepTr + LeRaC [21]	70.95	-	-
CopyPaste + Noise [2]	-	-	75.87
TRILL (10-folds) [20]	-	75.10	75.17
Ours (*EMix-NS*)	**77.25**	**77.26**	**77.06**

5. CONCLUSIONS

This paper introduces EMix - a simple but effective data augmentation method for speech emotion recognition. EMix creates new data samples by linear interpolation of selective data pairs from the emotion dataset. The key idea lies on its data selection schemes: mixing with *Neutral* samples and mixing within the same emotion. EMix is label-preserving, has no hyperparameter, so it is easy to use, and save training time. We develop a transformer-based network to validate the efficacy of EMix. Experiments on the two public datasets demonstrate its superiority over all other augmentation methods. Our proposed system also achieves competitive results in comparison to state-of-the-art methods.

6. REFERENCES

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